

Machine Learning- Introduction

What is Machine Learning?

- Machine Learning (ML) enables systems to learn patterns from data
- Improves performance without explicit rule-based programming

Why ML in Security?

- Cyber attacks evolve faster than manual rules
- ML adapts automatically to new threats
- Reduces dependence on static signatures

Security Example

- ML-based IDS learns:
 - Normal network traffic
 - Malicious traffic behavior
- Detection accuracy improves over time

Types of Machine Learning

1. Supervised Learning

- Training data contains **input + correct output labels**
- Learns a mapping from input → output

Security Example

- Email labeled as *spam* or *not spam*
- Malware files labeled as *malicious* or *benign*

Types of Machine Learning

2. Unsupervised Learning

- Training data has **no labels**
- Model discovers hidden patterns or structures

Security Example

- Identifying unusual network traffic
- Detecting insider threats without prior attack labels

Types of Machine Learning

3. Semi-Supervised Learning

- Uses **small labeled data + large unlabeled data**
- Common in security due to lack of labeled attacks

Security Example

- Few labeled malware samples, thousands of unlabeled binaries

Sub-Categories of Machine Learning Methods

Supervised Learning

1 Classification

- ▶ Discrete Class Label

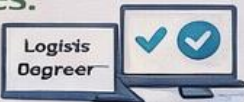
✓ Classification

- Predicts yes / No
- Attack / Normal
- Malware / Benign



Security Examples:

- ▶ Spam detection
- ▶ Intrusion detection
- ▶ Random Forest | SVM | Neural Networks



✓ Regression

- No Labels, Finds Groups
- Risk Score
- Probability of Attack
- Severity Level



Unsupervised Learning

2 Clustering

- ▶ No Label, Finds Groups

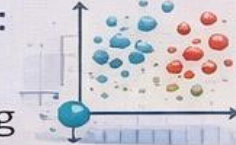
✓ Anomaly Detection

- & Rare / Abnormal Patterns
- Intrusion/ Insider/ IoT Anomaly Detection



Security Examples:

- ▶ K-Means | DBSCAN
- ▶ Hierarchical Clustering



✓ Dimensionality Reduction

- ▶ Fewer Features, Easier Learning
- Feature Reduction / Attack Visualization
- ▶ PCA / t-SNE / Autoencoders



Semi-Supervised Learning

3 Self-Training

- ▶ Few Labels, Retrain from New

✓ Co-Training

- Learns From Multiple Views
- Traffic + Log Based Detection



Security Examples:

- ▶ Self-training SVM
- ▶ Psuedo-labeling



✓ Graph-Based

- ▶ Propagates Labels in Graphs
- ▶ Botnet / Malware Graphs
- ▶ Semi-Supervised GANs / VAES



✓ Supervised: Classification, Regression 🔍 Unsupervised Learning ⚡ Semi-Supervised Learning

🟢 Used when 🟢 Labels are available 🔍 Discover & Semi-Supervised Learning ➤ 🔍 Predict + Discover 📍 Summaries

Supervised Learning

A. Classification

- **What it does**
 - Predicts a **discrete class label**
- **Typical Outputs**
 - Yes / No
 - Attack / Normal
 - Malware / Benign
- **Security Examples**
 - Spam detection
 - Intrusion detection
 - Malware family classification
- **Common Algorithms**
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - SVM

Supervised Learning

B. Regression

- **What it does**
 - Predicts a **continuous numeric value**
- **Typical Outputs**
 - Risk score
 - Probability of attack
 - Severity level
- **Security Examples**
 - Attack risk scoring (0–1)
 - Alert severity estimation
 - Likelihood of compromise
- **Common Algorithms**
 - Linear Regression
 - SVR
 - Neural Networks

Unsupervised Learning

A. Clustering

- **What it does**

- Groups similar data points together
- No labels required

- **Security Examples**

- Network traffic grouping
- User behavior profiling
- Zero-day attack discovery

- **Common Algorithms**

- K-Means
- DBSCAN
- Hierarchical Clustering

Unsupervised Learning

B. Dimensionality Reduction

- **What it does**
 - Reduces number of features while preserving information
- **Security Examples**
 - Feature reduction before IDS training
 - Visualization of attack patterns
- **Common Algorithms**
 - PCA
 - LDA
 - Autoencoders

Semi-Supervised Learning

A. Self-Training Methods

- **What it does**

- Train model on labeled data
- Predict labels for unlabeled data
- Iteratively retrain

- **Security Example**

- Few labeled malware samples + many unknown binaries

- **Algorithms**

- Self-training SVM
- Pseudo-labeling

Semi-Supervised Learning

B. Co-Training

- **What it does**

- Uses multiple views of data
- Models teach each other

- **Security Example**

- Network traffic + system logs combined

Semi-Supervised Learning

C. Graph-Based Methods

- **What it does**
 - Uses relationships between data points
 - Propagates labels through a graph
- **Security Example**
 - Malware similarity graphs
 - Communication graphs in botnets